Exercises week 2

8CM10 - Introduction to Quantum Physics and Structure of Matter

Compton Scattering

1. A photon with 10 keV energy collides with a free electron at rest and is scattered through an angle of 60 deg. Compute:
    a. The change in energy and frequency of the photon
    b. The kinetic energy, momentum and direction of the electron

2. X-ray photons of wavelength 0.02480 nm are incident on a target and the Compton-scattered photons are observed at 90.0◦.
    a. What is the wavelength of the scattered photons?
    b. What is the momentum of the incident photons? Of the scattered photons?
    c. What is the kinetic energy of the scattered electrons?
    d. What is the momentum (magnitude and direction) of the scattered electrons?

3. Gamma rays of energy 0.662MeV are Compton scattered.
    a. What is the energy of the scattered photon observed at a scattering angle of 60.0°?
    b. What is the kinetic energy of the scattered electrons?

De Broglie Wavelengths

4. By doing a nuclear diffraction experiment, you measure the de Broglie wavelength of a proton to be 9.16 fm.
    a. What is the speed of the proton?
    b. Through what potential difference must it be accelerated to achieve that speed?

5. Find the potential difference through which electrons must be accelerated (as in an electron microscope, for example) if we wish to resolve:
    a. a virus of diameter 12 nm
    b. an atom of diameter 0.12 nm
    c. a proton of diameter 1.2 fm.

Uncertainty Principle

6. Suppose a traveling wave has a speed v (where $v = \lambda f$). Instead of measuring waves over a distance Δx, we stay in one place and count the number of wave crests that pass in a time Δt. Show that $\Delta f \Delta t \sim \epsilon$ is equivalent to $\Delta x \Delta \lambda \sim \epsilon \lambda^2$ for this case (see 4.3 in Krane).

7. The lifetime of an excited state is 10^{-8}s. What is the minimum uncertainty in
    a. The energy
    b. The frequency

8. The speed of an electron is measured to within an uncertainty of $2.0 \cdot 10^4\ \text{m/s}$. What is the size of the smallest region of space in which the electron can be confined?